MILITARY SPECIFICATION

TRUCKS, PALLET, POWERED, ELECTRIC, LOW-LIFT, STAND-UP RIDER TYPE

General Specification For

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers general requirements for electric powered, low-lift, stand-up rider pallet trucks with forks.

1.2 Classification. Pallet trucks shall be one of the following types and sizes (see 6.2):

Type E - Non-spark enclosed

Type EE - Spark enclosed

Size I - 4,000 (1815 KG) pounds capacity

Size II - 6,000 (2722 KG) pounds capacity

Size III - 10,000 (45351 KG) pounds capacity

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Navy Ships Parts Control Center, Post Office Box 2020, Attention: Code 0302, Mechanicsburg, Pennsylvania 17055 by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

SPECIFICATIONS

Federal

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QQ-C-320 - Chromium Plating (Electrodeposited)

TT-C-494 - Coating Compound Bituminous Solvent Type Acid Resistant

TT-E-489 - Enamel, Alkyd, Gloss (For Exterior and Interior Surfaces)

O-E-915 - Extinguishers, Fire, Dry-Chemical (Hand-Portable)

Military

MIL-P-116 - Preservation, Methods of

MIL-V-173 - Varnish, Moisture-and Fungus-Resistant (For the Treatment of Communications, Electronic, and Associated Electrical Equipment)

MIL-T-704 - Treatment and Painting of Material

MIL-F-3541 - Fittings, Lubrication

MIL-G-3859 - Grease Gun, Hand, High Pressure, Lever-Operated, Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity)

MIL-M-3971 - Meter, Time Totalizing, Non-Hermetically Sealed, Electric, AC and DC

MIL-M-21861 - Manual, Technical, Materials Handling Equipment, Preparation, Contents and Approval STANDARDS

Federal

FED-STD-H28 - Screw Thread Standards For Federal Services

FED-STD-595 - Colors

Military

MIL-STD-105 - Sampling Procedures and Table for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-130 - Identification Marking of U. S. Military Property

MIL-STD-162 - Materials Handling Equipment, Preparation for Shipment, Storage, Cyclic Maintenance, Routine Testing and Processing

MIL-STD-461 - Electromagnetic Interference Characteristics: Requirements for Equipment

MS-15367 - Battery, Storage, for Electric Powered Industrial

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Trucks and Tractors

MS-16966 - Tire, Cushion, Permissible Sizes and Loading for Use on Original Material Handling Equipment

MS-16967 - Tire, Solid, Permissible Sizes and Loading for Use on Original Material Handling Equipment

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall govern.

AMERICAN NATIONAL STANDARDS INSTITUTE

Safety Standard For Powered Industrial Trucks (ANSI B56.1)
Safety Standard For Electric-Battery-Powered Industrial Trucks (ANSI B56.3)

(Application for copies should be addressed to: American National Standards Institute, 1430 Broadway, New York, New York 10018).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

Book of ASTM Standards

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION

Code Of Federal Regulations
29 CFR, CHAPTER XVII Part 1910
Occupational Safety and Health Standards

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402).

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

NEMA Standards

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, New York 10017).

SOCIETY OF AUTOMOTIVE ENGINEERS, INC.

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, Pennsylvania 15096).

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UNDERWRITERS' LABORATORIES, INC.

Standards for Safety for Electric Battery Powered Industrial Trucks UL 583 (Application for copies should be addressed to: Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, New York 11746; 333 Pfingston Road, Northbrook, IL 60062; or 1655 Scott Boulevard, Santa Clara, California 95050).

3. REQUIREMENTS

3.1 Standard product. The electric powered, low-lift, stand-up rider type pallet truck with forks (hereinafter called truck), shall be constructed and equipped to meet all requirements specified herein and shall be complete and ready for operation where delivered. The end items and major and minor assemblies shall be new and of a standard design or model product used by the end item manufacturer current at time of bidding, except for such minor modifications from manufacturer's standard as may be required to conform to these specification requirements. There shall be no change in components to offset interchangeability within individual contract items without authorization by the contracting officer. The component parts of the unit need not be the products of the same manufacturer; however, the manufacturers shall be so established in the industry that prompt and continuing service and delivery of replacement repair parts may be assured.

3.1.1 First article testing and approval. Unless otherwise specified (see 6.2), the supplier shall furnish a truck within the time frame specified in the contract or purchase order to prove, prior to starting production, that his production methods and choice of design detail will produce trucks that comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the preproduction model during production shall be subject to the approval of the contracting officer. Approval of the preproduction model by the activity concerned shall not relieve the supplier of his obligation to furnish trucks conforming to this specification. The approved preproduction model may be presented for acceptance as the last unit for delivery under the contract after being rehabilitated to reflect a new appearance. This shall include the furnishing of new tires.

3.1.1.1 Initial production. The supplier shall furnish to the Government one truck for testing as specified in paragraph 4.4.

3.1.2 Certificates of compliance. When specified herein, or upon request, a certificate of compliance with the requirements of this specification shall be

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furnished. Such certificates shall include backup data which shall be made available for review. The certificate of compliance shall read as follows:

> This certifies that the _____(component, part or assembly) complies with all the provisions and requirements of Specification _____paragraph_____.

The certificates shall be signed by a duly authorized agent of the contractor.

3.2 Materials. Materials not definitely specified shall be at least equal to the end item manufacturer's normal commercial product, and shall meet test requirements specified in section 4. "Normal commercial product" shall be interpreted to mean an item furnished by an end item manufacturer normally engaged in commercial production of the item(s) covered by this specification. The Government shall require any bidder, who proposes to furnish other than a "normal commercial product" or Government specified item, to submit, prior to and as a condition precedent to the award of contract, detailed engineering drawings of those parts he proposes to furnish in lieu thereof for approval of the contracting officer.

3.2.1 Castings. Castings shall be of the quality normally used in commercial practice, free from blowholes, porosity, shrinkage, cracks, or other defects. Castings shall not be repaired that adversely affect the strength or intended use. Sufficient finish shall be allowed to permit machining for leak-free sealing at covers, closure points, and fasteners.

3.2.2 Plating. All threaded fasteners, washers and cotter pins required to fabricate the truck, except those designed to be in contact with oils in reservoirs, shall be zinc coated or shall be protected by a method of equal corrosion resistance.

3.2.3 Screw threads. Screw threads shall be in accordance with FED STD H-28.

3.3 Safety. The truck shall conform to the applicable requirements for Underwriters' Laboratories, Inc., "Standards for Safety for Electic Battery Powered Industrial Trucks (UL 583)" for type E or type EE trucks as applicable OSHA 29CFR, Chapter XVII, Part 1910, and the applicable requirements of Safety Standard for Powered Industrial Trucks (ANSI B56.1) and Standard for Electric Battery Powered Industrial Trucks (ANSI B56.3).

3.3.1 Certification. The contractor shall submit to the contracting officer or his authorized representative satisfactory evidence that the trucks furnished under this specification meet or exceed the requirements of UL 583 Type EE. Acceptable evidence of meeting these requirements shall be certified test reports from recognized independent testing laboratories acceptable to the government, indicating that the trucks conform to the requirments of UL 583.

3.4 Maintainability. Provision shall be made for ready adjustment, servicing, or replacement of all electrical assemblies and components, battery, wearing parts of lift mechanism, brake and components, wearing parts of steering assembly, tires,wheels, and horn without the removal of any major parts other than covers. If hand access openings are used on the exterior of the truck, the edge of the opening shall be smooth and shall be provided with a removable or hinged cover, whenever it is required. There shall be no interference to

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the servicing or draining of oils and lubricants to or from any assembly or components by frame members or other obstructions, and without requiring the use of flexible connections or other special device. Draining plugs shall provide for complete drainage. Compliance with specific maintenance requirements for the equipment shall be timed in accordance with 4.6.17. There shall be no part of the hydraulic system extending below the frame of the truck unless it is properly guarded against damage.

Trucks shall bear either the Underwriters Laboratories or an independent testing laboratory label indicating their stamp of approval for the EE rating.

3.4.1 Maintenance operations. Unless otherwise specified herein, each operation in the following maintenance operations list shall be accomplished by one man in not more than the time specified, using common tools and special tools, if any, furnished with the truck.

Maintenance Operation (Time in Minutes)	E	EE
a. Remove and Replace Drive Motor Brushes	60	90
b. Remove and Replace Hoist and Tilt Motor Brushes	60	90
c. Remove and Replace Steer Motor Brushes	60	90
d. Remove and Replace all Contactor Tips	60	90
e. Remove and Replace all Filters, Screens and Strainers in Hydraulic System	105	105
f. Bleed and Adjust Brakes and Refill Master Cylinder (2 men) (4-wheel brakes excepted)	45	45

3.4.2 Power consumption. When specified (see 6.2), the truck shall be tested for compliance with the power consumption requirements in accordance with the power consumption requirements of 4.6.16.

3.5 Design and construction. These trucks shall be designed for the rated load as listed in Table I and the specified lift height in accordance with 3.6.2. Design of trucks covered by this specification and all components therein, shall insure safe, efficient, and economical operation under normal service conditions. There shall be no evidence of accelerated wear, or failures, or permanent deformation when operated under normal service conditions or when tested as specified in section 4. There shall be no exposed bolts, clamps, gages, fittings, or other appendages which can be caught or hooked while the truck is working in a confined space.

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TABLE I

Design Capacity

Size	Rated Loads in Pounds	At Load Center
I	4000 (1815 KG)	24-Inch (610 MM)
II	6000 (2722 KG)	24-Inch (610 MM)
III	10000 (4535 KG)	24-Inch (610 MM)

3.5.1 Electrical Equipment.

3.5.1.1 Resistors. Resistors when used shall be rated for continuous use and shall not burn-out or fuse when tested in accordance with this specification. For type EE trucks, the use of resistors shall be avoided to the maximum extent practical.

3.5.1.2 Enclosure and shields - Type EE Trucks. Type EE trucks shall comply with the requirements of UL 583. Motor enclosures shall be of the totally enclosed type. Electrical controls shall be completely enclosed with no ventilation openings permitted. A UL or Factory Mutual label attached to the truck with type designation will be evidence of compliance.

3.5.1.3 Overcurrent devices. In addition to the requirements of 3.3, overcurrent devices shall be provided to properly protect the power and control circuits. The devices shall be sized to carry the rated current of the battery at the one hour rate or a maximum current drain when traveling or lifting whichever is greater.

3.5.1.4 Resistor housings, heat-dissipating, and heat-actuated devices --Type EE Trucks. The outside surface temperature of resistor housings shall stabilize at 250 degrees Fahrenheit (degrees F) (120 degrees C), or below when truck is operating continuously in the speed which creates the most heat in a minimum ambient temperature of 95 degrees F (35 degrees C). The housing may be cooled by heat-dissipating methods such as oil coolant, fins or other mechanical means to aid in the dissipating of heat. When oil coolant is used there shall be no leakage from oil containers and oil shall not contact control resistors. Not less than two safety thermal relays (heat-actuated devices) shall be mounted on the housings at critical heat areas and shall be set to instantaneously disconnect the battery power to the affected circuit(s) if the housing temperature exceeds 250 degrees F (120 degrees C). (Critical heat areas are those surfaces which exceed 200 degrees F (93 degrees C) during the course of testing required by this specification.) There shall be no evidence of failure to meet these requirements when tested as specified in 4.6.14. A certificate of compliance is required.

3.5.1.5 Static discharge device. Type EE trucks shall be equipped with two static tires to discharge static electricity through contact with the road surface.

3.5.1.6 Conductors. For Type EE trucks all conductors, except battery leads emerging from enclosure(s), shall be through fittings, so that sparks shall not

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propagate to the atmosphere and dust shall not enter from the atmosphere into the enclosure(s). Identification of control leads shall be provided through color coding, numbered markings, tags, or similar means to identify conductor runs.

3.5.2 Motor and motor accessories

3.5.2.1 Motors - Type E trucks. Traction motors shall be industrial type with a high starting torque designed for efficient operation on 24 direct current voltage. Motors shall be capable of withstanding the stresses and current loads induced by tests as required by this specification. The resultant strains and temperature rise shall not cause failure of any component part or excessive pitting other than normal discoloration of the commutators. The hydraulic pump motor shall successfully withstand the stresses and current loads induced by tests required by this specification. Provision shall be made that all power to the traction motor is interrupted when the operator leaves the truck.

3.5.2.2 Motors - Type EE trucks. In addition to the requirements of 3.5.2.1, motor(s) for type EE trucks shall be provided with one or more thermal relay switches which shall cause disconnection of battery power to the affected circuit before the outside surface temperature of the motor(s) exceed 250 degrees F (120 degrees C). Temperatures shall stabilize at 250 degrees F (120 degrees C) during specified test in 4.6.14. Thermal relays shall be located in critical heat areas and shall interrupt the current only to the over-heated circuit.

3.5.2.3 Insulation. Insulation for motors shall be Class B, F, or H as defined in NEMA Standards except the insulation shall not contain silicone materials. Silicone material in any form shall not be used in the motor. Insulation shall be selected to give the best balance of overall operating characteristics of torque, speed, and temperature rise. The class of insulation shall be determined by the material with the lowest temperature rise used in the insulation system. Armature and fields need not have the same class of insulation, but each part shall meet the temperature rise of its class of insulation. The electrical and mechanical properties of the insulated wiring shall not be impaired by the application of the temperature permitted for Class H material (the word "impaired" is here used in the sense of causing any change which could disqualify the insulating material for continuous service).

3.5.2.4 Speed controller. The speed control shall be either resistor type or a silicone controlled rectifier (SCR). The control shall be of heavy duty design for continuous use and shall not incur failure or damage when tested in accordance with Section 4. If a resistor control is provided, it shall provide not less than 3 speeds forward and 3 speeds reverse through a stepping mechanism. The controller shall provide means to permit precision maneuvering in restricted areas. It shall provide smooth acceleration between each successive speed range, and shall have automatic electrical or mechanical interlocks to prevent direct connection into high speed, and shall have means to prevent starting in other than first speed. The controller shall have automatic interlocks to prevent reversing except in first speed. All contactors on a given truck shall be designed to provide maximum interchangeability of parts. Contactors shall be designed for operation on 24

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volts direct current and shall have easily replaceable tips of copper or silver alloy composition. Replacement of contactor tips shall be accomplished by use of hand tools only. There shall be no evidence of failure in any part of the controller when the truck undergoes tests specified herein. Speed controller may be of the oil-immersion type.

3.5.3 Battery and battery accessories.

3.5.3.1 Battery. Unless otherwise specified (see 6.2), batteries will be furnished by the Government at destination. Unless otherwise specified (see 6.2) battery will be 24 volts and have a minimum capacity of not less than 725 ampere-hours. Unless otherwise specified (see 6.2) battery dimensions shall be 36-1/4 inches (920 mm) wide by 14-1/4 inches (362 mm) deep by 31 inches (787 mm) high with dimensional tolerances of plus 0 inches minus 1/8 inch (3 mm). Batteries will have covers when used in trucks having battery wells (see 3.5.3.3.2).

3.5.3.2 Battery connectors. Unless otherwise specified (see 6.2) trucks shall be equipped with 2 contact type EC battery connectors. Battery leads shall be in accordance with MS-15367.

3.5.3.3 Battery compartment or well. The truck shall be supplied with a battery compartment or well as specified (see 6.2).

3.5.3.3.1 Battery compartment. The battery compartment shall be constructed to accommodate the batteries described in paragraph 3.5.3.1. Battery compartment or sides of support shall not interfere with operator's line of sight to the tips of the forks when he is in the operating position. For type EE trucks battery compartment shall be of spark enclosed construction. The inside of the compartment shall be coated in accordance with TT-C-494 type II, coating compound, acid resistant. On type EE trucks, a provision shall be made for locking the compartment with a padlock furnished by the Government at destination.

3.5.3.3.2 Battery well. The battery well shall have sides 6 inches minimum height or shall have a retainer approximately half way up the battery. Construction shall be such that the battery specified in 3.5.3.1 will fit securely in the well and can be serviced without removal from the truck.

3.5.4 Drive unit. Each gear (with the exception of worm gears, if used), sprocket, shaft, and axle, shall be of high grade treated alloy steel. The drive shall be a gear reduction unit(s) connected directly to the driving motor(s), totally enclosed and operating in a constant oil bath. Drive gear housing(s) shall be equipped with filler and magnetic drain plug. All required drain plugs, check plugs, or oil level checking and filling device in the transmission shall be readily accessible without removing major components. With drive wheels locked, the drive assembly shall be capable of withstanding a static test load applied to the input of the transmission without failure, fracture or deformation of any component. The test torque shall be calculated as follows: $T = .04WD/R$ ($T = .34WD/R$).

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In which:

T = Test torque in foot-pounds (Joules).

W = Combined weight of vehicle and load in pounds (kilograms).

D = Diameter of drive wheels in inches (millimeters).

R = Overall gear ratio of the drive system.

When specified (see 6.2), complete drive assembly may be tested prior to or after complete assembly of the truck at option of the contractor in accordance with test as specified in paragraph 4.6.18.

3.5.5 Hydraulic system. Power used for lifting shall be furnished by a hydraulic pump directly driven by a motor independent of the traction motor. It shall be designed and constructed to automatically hold a rated load in any position upon release of the lowering mechanism. Mechanisms used in raising and lowering the forks shall be protected from damage. The hydraulic reservoir shall be provided with an easily accessible filler plug for filling and checking the fluid level and shall have sufficient capacity to fill the hydraulic system plus 30 percent reserve fluid. A pressure relief valve shall be provided which shall open at 110 to 120 percent of the rated load to protect the hydraulic system from damage due to overload and continued operation of the lift mechanism after the load is lifted to maximum height.

3.5.5.1 Piston rods. Piston rods for the hydraulic system shall be plated with an electro-deposited hard chrome-plate finish in accordance with QQ-C-320, class 2. Surfaces shall be thoroughly cleaned prior to plating in accordance with normal commercial practice.

3.5.5.2 Hydraulic fittings, hoses, and tubing. All components utilizing hydraulic fluid from the hydraulic reservoir excluding brakes and steering systems shall conform to the following standards. Hydraulic fittings used on valves and cylinders shall be the "O" ring straight thread type in accordance with SAE J514 but not limited to the fittings listed therein. When hydraulic tubing is used, fittings shall conform to SAE J514 37 degrees two or three piece tube assembly. All other fittings shall be of the 4 bolt split flange type in accordance with SAE J518 or the "O" ring straight thread type except for barbed or clamp types at both ends of suction, return, and cylinder bleeder lines, and dry seal connectors on lines where hydraulic pressure does not exceed 200 psi. Hose fittings shall be of the reusable screw-on type, 37 degree flared type female in accordance with SAE J516, except for the hydraulic hose flanged "O" ring connections which shall conform to SAE J519. Hydraulic hose sizes shall be restricted to the following sizes: 1/4", 5/16", 3/8", 13/32", 1/2", 5/8", 3/4", 1", 1-1/4", 1-1/2", 2", and 2-1/2". Hydraulic hose specification with respect to applications shall conform to the following:

1/4" - 3/4"	Suction and discharge	Double cotton braid SAE 100R3
1" - 2-1/2"	Discharge	Double cotton braid SAE 100R3 or equivalent

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3/4" - 2-1/2"	Suction	SAE 100R4
1/4" - 1/2"	Pressure	SAE 100R1 or SAE 100R5
1/2" - 1"	Pressure	SAE 100R2

3.6 Structure.

3.6.1 Chassis and frame. The frame and related structure shall be fabricated from structural steel and steel castings at least equal to the size and design of manufacturer's normal commercial frame members and shall be capable of withstanding 200 percent of rated load without permanent deformation. Frame members, bracings, and all their joints shall provide a rigid unit structure. The frame and related structure shall have sufficient depth to afford protection for working parts and shall be capable of withstanding, without permanent deformation, stresses induced by capacity loads and test in section 4.

3.6.2 Forks. Steel forks shall be provided and constructed from formed or fabricated channel cross sections capable of withstanding stresses induced by carrying out all applicable provisions of section 4. Fork length shall be 39-1/2 inches (1003 mm) to 40 inches (1016 mm) or 47-1/2 inches (1206 mm) to 48 inches (1220 mm) as specified (see 6.2). Maximum distance between outside edges of forks shall be 27 inches (686 mm). The space between the inside edges of the forks shall be 8-3/4 inches (222 mm) within a tolerance of plus 1/2 inch (12 mm) and minus 1/4 inch (6 mm). Lowered fork height shall not exceed 3-3/8 inches (86 mm). A minimum vertical lift of 4 inches (102 mm) shall be provided. Minimum underclearance, with forks lowered, shall be 7/8 inch (22 mm). Forks shall be horizontal within a tolerance of plus 1/16 inch (2 mm) and minus 1/8 inch (3 mm) at the ends. Manufacturing tolerances for forks shall be plus or minus 1/16 inch (2 mm). Forks shall be end tapered on the underside to facilitate entry into and withdrawal from double faced pallets. Raising and lowering mechanism shall be recessed, or guarded to prevent damage from impact and abrasion experienced during normal operation.

3.6.3 Pallet positioner. When specified (see 6.2) the truck shall be equipped with a manually operated pallet positioner or stop in order to properly position the 40 inch (1016 mm) length of a 40 inch (1016 mm) by 48 inch (1219 mm) pallet on 48 inch (1219 mm) forks.

3.6.4 Bearings. Rotating parts shall be mounted on anti-friction bearing, except as specifically indicated herein, to the extent normally provided in standard commercial practice and good design. Bearing load capacity, including applicable safety and life factors, as recommended by the bearing manufacturer, shall be equal to or greater than maximum bearing loading under normal truck operation.

3.6.5 Drive wheel(s) configurations. Trucks shall be equipped with one or two drive wheels. When one drive wheel is provided, one or more caster follow wheels shall be provided. The entire system shall be designed for laterally articulated mounting to maintain drive wheel contact with uneven floors.

3.6.5.1 Drive wheel(s). Wheel(s) shall be of the flat face or demountable type made from steel or cast iron mounted on ball or tapered roller bearings.

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Tapered roller bearing shall provide means for adjustment to compensate for wear. Readily accessible grease fittings shall be provided to facilitate lubrication in accordance with good automotive practice where sealed type bearings are not provided.

3.6.5.2 Tires. The drive wheel tire shall have nondirectional tread, molded onto a steel band, for pressing onto a flat faced wheel. The tire shall be new and unused and one of the sizes listed in MS 16966 or MS 16967. Tires may be static discharge type to meet the requirement of paragraph 3.5.1.5.

3.6.6 Caster wheels. Caster wheels, if used, shall have polyurethane tires and shall be designed and mounted to provide stability to the truck.

3.6.7 Load wheels. Trucks shall be furnished with 2 or 4 polyurethane load wheels, equipped with angular contact ball, tapered or spherangular roller bearing of the type best suited for the purpose recommended by the bearing manufacturer. Tapered roller bearings shall provide means for adjustment to compensate for wear. Wheels shall be positioned to function unobstructed with a maximum 6 inch (152 mm) end board and a minimum 7-1/2 inch (190 mm) wheel opening. Load wheels may have static discharge type tires to meet requirements of paragraph 3.5.1.5.

3.6.8 Drive unit mounting. When two drive units are used, mounting shall be in such a manner to insure articulation of the drives and insure contact with the floor by one drive wheel when the other wheel negotiates a 1-inch (25 mm) high obstacle block.

3.6.9 Steering. Steering shall be accomplished by a hand steering wheel equipped with steering knob. Wheel shall be located for convenient left hand operation when facing load. It shall be located to make it unnecessary for any part of the driver's body to project outside the plan outline of the truck. The steering mechanism shall turn the drive wheels through gear reduction or commercial equivalent. Chains, if used in the steering mechanism, shall be easily adjustable in an amount of not less than one complete link without removal of any truck structure or major component. Clockwise rotation of the horizontal steering wheel shall steer the truck to the right (operator facing load). The steering wheel shall turn the drive wheel(s) assembly at least 90 degrees in each direction from the neutral position. Steering handle in lieu of steering wheel may be furnished where this method is the only type normally used by a manufacturer of pallet trucks of this type. The use of steering handles in lieu of steering wheels must be approved by the contracting officer prior to contract award.

3.6.10 Brakes. Brakes shall be of the mechanical hydraulic or mechanical external contracting, internal expanding or disc type, mounted on the traction motor shaft, countershaft or drive wheel. Brakes shall be protected to insure satisfactory operation and shall provide for ready and immediate servicing and adjustment. Brake linings shall be of the molded type cement bonded to the brake shoes, except linings for disc brakes, which shall be of the conventional type bolted on the carrier plate. The brake system shall be interlocked with a relay switch to cause disconnection of the electrical circuit from the traction motor when the brakes are applied. Brakes shall be actuated by a foot pedal located for convenient left foot operation. Application shall be by raising the foot. Brakes shall be automatically applied when the operator leaves his

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normal operating position. Brakes shall be capable of retaining the truck with rated load on a 15 percent grade.

3.6.11 Body. The truck body shall be constructed of heavy steel sheet or plate equal to that used in the manufacturer's normal commercial product. The truck body shall completely cover the drive unit assembly, steering mechanism and speed controller, when these components are not housed in individual containers. Provisions shall be made for immediate and easy access to all working parts requiring inspection or maintenance. Covers shall be secured with bolts or quick opening fasteners removable with ordinary hand tools.

3.6.11.1 Operator's platform. The operator's platform shall be fabricated from steel plate with non-skid rubber tread not less than 1/2 inch (12 mm) thick cement bonded thereon. Operator's platform guard shall be designed for rapid and unobstructed egress from the rear of the truck and shall be strong enough to prevent its collapsing and pinching the operator. The operator's platform shall have sufficient depth to permit the operator to stand within the plan outline of the truck. Platform guard shall be constructed to withstand a compression load equal to the weight of the loaded vehicle applied in the direction of travel (see 4.6.12).

3.7 Controls and instrumentation. Where applicable, the provisions of UL 583 shall apply. Electric controls and instrumentation shall be constructed to be moisture and weather resistant. On type EE trucks, all instruments shall be of spark-enclosed construction or mounted in a spark-enclosed enclosure. All controls shall be provided with position markings (decals are not acceptable) at the control(s).

3.7.1 Directional speed controls. Direction and speed controls, push buttons, levers, or switches shall be mounted in the operator's compartment and shall be conveniently located for right hand operation. Means shall be provided to prevent simultaneous actuation of forward and reverse controls. Rotating handles shall be arranged so that forward rotation will cause forward motion of truck and vice versa. Trigger speed control, if used, shall be so arranged that depressing the trigger causes increase in speed in the direction predetermined by the direction selector switch. If buttons or switches are used, the one located nearest the operator shall, when actuated, cause forward motion of the truck and vice versa. All controls shall be so constructed that they return to neutral position when released.

3.7.2 Lift and lowering controls. The lifting and lowering controls shall be a switch, lever, or button. It shall be conveniently located for operation by the right hand. If buttons or switches are used for both lifting and lowering, the one located nearest the operator or the upper, if arranged vertically, shall, when actuated, raise the forks. The lift control shall be so arranged that the forks will remain at any position reached when the control is released.

3.7.3 Instruments. All gages and meters shall be products of manufacturers regularly engaged in producing these types of instruments. They shall be flush mounted in a location convenient to and plainly visible to an observer standing on the floor.

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3.7.3.1 Hour meter. An electric hour meter, non-braked, flush mounted, direct reading type, conforming to MIL-M-3971 type I and UL583 shall be furnished. The meter shall register the total number of hours of operation of the traction and pump motors.

3.7.4 Horn. Truck shall be equipped with an electric horn with push button mounted for convenient hand operation. The horn shall be operable without requiring the operator to release the steering wheel or speed control.

3.7.5 Fire extinguisher. When specified (see 6.2), each truck shall be equipped with a 4 pound (2 KG) internally operated dry chemical charged fire extinguisher with rechargeable cylinder having a minimum UL rating of 6BC. When specified (see 6.2), each truck shall be equipped with a fire extinguisher in accordance with Federal Specification O-E-915, type I, class 2, size 2-1/2 with a minimum UL rating of 8B:C.

3.8 Performance.

3.8.1 Maximum load. The frame, forks, and related structure shall be capable of withstanding, without permanent deformation or failure, 200 percent of rated load with forks elevated to maximum lift height when tested in accordance with 4.6.1.

3.8.2 Shock load. When loaded with 150 percent of rated load, the truck shall withstand without permanent deformation, failure, or excessive settling, the shock test described in 4.6.2.

3.8.3 Pallet entry and withdrawal. The truck shall be capable of complete entry into and withdrawal from any empty double faced standard military pallet without moving the empty pallet more than 6 inches (152 mm) in either direction or dragging the ends of the forks along the bottom face of the top boards (see 4.6.4).

3.8.4 Turning radius. Trucks shall be capable of turning 360 degrees in either direction within a circle. For Size I and Size II trucks, the radius shall not be greater than 73 inches (1855 mm) with 40 inch (1016 mm) forks and 81 inches (2058 mm) with 48 inch (1220 mm) forks. For Size III trucks, the radius shall not be greater than 90 inches (2286 mm) with 40 inch (1016 mm) forks and 98 inches (2490 mm) with 48 inch (1220 mm) forks. All parts of truck structure shall remain within this circle when tested in accordance with 4.6.3. The truck shall meet this requirement within 3 trials.

3.8.5 Speed. While traveling on a smooth, dry concrete level surface with rated load, Size I, Size II, and Size III trucks shall be capable of forward and rearward speeds of not less than 4-1/2 miles per hour (7.2 KM/HR), 4 miles per hour (6.4 KM/HR), and 3 miles per hour (4.8 KM/HR) respectively, measured in accordance with 4.6.5.

3.8.6 Slope ascension. Trucks shall be capable of ascending a 15 percent grade on a dry concrete surface when carrying rated load in forward direction. In addition, trucks shall be able to accelerate from a dead stop on this slope carrying rated load as described in 4.6.6.

3.8.7 Lifting operation. The lift mechanism shall be capable of raising and

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lowering double faced standard military pallet with rated capacity loads. With wheels in the recessed position and forks fully engaged into the pallet, there shall be 1/4" (6 mm) minimum clearance between the bottom of the pallet upper deckboards and any part of the truck that could physically contact it. Contact between the pallet and the raising and lowering mechanism shall not be made in the elevated position.

3.8.8 Lifting and lowering speed. Time required to raise rated load from lowered position to maximum lift height shall not exceed 6 seconds. Lowering speed of forks shall be hydraulically controlled.

3.8.9 Lift assembly. The lift assembly shall be capable of completing 500 endurance cycles as described in 4.6.11 with no signs of appreciable wear. "Appreciable wear" is defined as any apparent inability to lift and hold rated load with not more than 1/2 inch (12 mm) drift per minute.

3.8.10 Hydraulic system. The hydraulic system shall show no leaks or breakage when truck is subjected to test prescribed in 4.6.9.

3.8.11 Brakes. The brakes shall be capable of stopping the truck carrying rated load from maximum speed within a distance of 5 feet (152 mm). Test shall be performed as prescribed in 4.6.10.

3.8.12 Stability. Stability shall be such that there is no overturning when truck is unloaded and when a laterally centered rated load is placed on forks and elevated to maximum height, speed is at maximum, and the truck maneuvers as prescribed in 4.6.8.

3.9 Electromagnetic interference control. Unless otherwise specified (see 6.2), the trucks shall be classified as class IIID, table 1, of MIL-STD-461. For Navy use, type EE trucks shall be classified as class IIIC except that the 20 decibel (db) relaxation of MIL-STD-461, RE05, does not apply.

3.10 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.10.1 Lubricants. The trucks shall operate as specified herein when lubricated with military lubricants as specified (see 6.2).

3.10.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.10.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

3.11 Fungus and moisture resistance. When specified (see 6.2), the electric circuitry, including all components and connections, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173 with the following exceptions:

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(a) Components of circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated. When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of .001 inch (.025 mm) to component or element surfaces previously cleaned and prepared to assure that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.12 Nameplate. The manufacturer shall provide and install a corrosion-resistant metal nameplate, according to STANDARD MIL-STD-130 (for Department of the Navy (less U. S. Marine Corps) stock numbers are not required) in a visible and safe location on the truck, with the following information indented or embossed on the plate:

TRUCK, PALLET, POWERED, ELECTRIC, LOW-LIFT, STAND-UP RIDER TYPE

CAPACITY . . . . . Pounds (Kilograms)

NAME OF MANUFACTURER . . . . . .

SERIAL NUMBER. . . . . .

MODEL NUMBER . . . . . .

NSN NUMBER (include only when specified) . . . . . .

REGISTRATION NUMBER. . . . . (month and year)

CONTRACT OR ORDER NUMBER . . . . . .

INSPECTED . . . . . (name of inspection office)

SERVICE WEIGHT . . . . . .(pounds) (kilograms) (less battery)

IDENTIFICATION OF MAINTENANCE MANUAL

. . . . . (truck serial number or other)

U. S. . . . . .

3.13 Registration data. The manufacturer shall execute registration data cards as required by the contracting officer. The assigned registration number for each truck shall be painted, by the contractor, in black enamel, in block numerals 3 inches (76 mm) high by 1/2 inch (12 mm) line width for letters and numerals and in a prominent place on truck body or battery compartment. (see 6.2).

3.14 Treatment and painting. The truck shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, except that the finish coat shall be yellow gloss enamel color No. 13538 of FED-STD 595 and conforming to TT-E-489.

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An alternate commercial rust-inhibiting primer may be used in lieu of the primer specified provided that the substituted primer has all the service properties of the specified primer and is compatible with the alkyd finish coat specified. When a preproduction model is required for test and evaluation, it shall be painted with one primer coat only. Machined or polished surfaces shall be masked prior to painting. Parts subject to temperatures over 450 degrees F (232 degrees C) shall be painted with heat-resistant paint in accordance with the manufacturer's standard practice. Lift chains, sprockets, rollers, plastics, fabric hoses (except hydraulic hoses) and drive belts shall not be painted except that slight overspray will be accepted, provided there is no interference with the proper functioning of the component. Paint must be kept off working surfaces where interference with working parts would result. Truck accessories, i.e., motor controls, instruments and wiring, plated or painted by the manufacturer of such items shall not be painted by the contractor and shall be adequately protected or masked from overspray.

3.15 Technical publications. Unless otherwise specified (see 6.2), one copy of the technical manual conforming to MIL-M-21861 shall be furnished by the manufacturer with each truck at the time of shipment. The manuals shall be placed in a protected place in the equipment. Each manual shall be preserved and packaged in a transparent plastic bag in accordance with Method IC-1 of MIL-P-116 and the bags heat sealed. Manuals shall be complete in every detail in accordance with good automotive practice and should conform to the requirements of the procuring activity, including complete detailed wiring diagrams of silicon controlled rectifiers (SCR) and control board(s), circuit boards, etc., where used. The use of advertising of any type in this manual shall be avoided to the maximum extent possible. Additional copies of the manual and other data requirements shall be furnished as specified (see 6.2).

3.16 Repair parts and maintenance tools. Such repair parts and maintenance tools as specified shall be furnished (see 6.2).

3.17 Items to be furnished with truck. No shipment of trucks shall be made unless technical manuals (see 3.15), repair parts, maintenance tools, operating tools, and accessories as required, are included with the shipment or unless approval for shipment of the trucks without such items has been received from the contracting officer.

3.18 Repair parts support. Contractor will be responsible for making available parts that are peculiar to the contractor or produced exclusively for him. A list of these parts shall be furnished the cognizant technical activity through the contracting officer as a part of the provisioning data to be submitted. These parts shall be available for a period of no less than ten years from date of contract. Should the contractor not make these parts available, he shall furnish the cognizant technical activity sufficient production design and manufacture processing data to facilitate procurement from another source. The contractor shall furnish a certificate that indicates compliance with the foregoing.

3.18.1 Provisioning. The Government in conjunction with the contractor will, as a part of provisioning, develop and establish anticipated usage and pertinent maintenance data of parts peculiar or produced exclusively for him.

3.19 Workmanship. The truck shall withstand any operation specified herein

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3.19 Workmanship. The truck shall withstand any operation specified herein without permanent deformation, breakage of connections, malfunction, or component interference caused by incorrect workmanship. All parts of the truck before and after painting shall be clean and free from sand, rust, dirt, fins, pits, sprues, scale, flux, and other harmful features. Edges and surfaces exposed to the operating and maintenance personnel shall be smooth and rounded to the extent a hazardous surface does not exist. Pivot points shall have hardened pins and bushed holes.

3.19.1 Steel and other metal fabrication. Metals used in the fabrication of the truck shall provide original quality surface finish and shall be free from kinks and sharp bends. Metals having eroded or pitted surfaces are not acceptable. The straightening of materials shall be done by methods that will not cause weakening or injury to the material. Burrs and sharp edges in holes and on sheets, plates, and members shall be removed sufficiently to assure correct fits and to prevent loosening of fasteners and damage to components. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Splatter shall be removed from exposed cuts and from reentrant cuts. Heated metals shall be allowed to cool slowly, and overheating shall be avoided in accordance with the recommendations of the metal manufacturer. All bends of a major character and all modular assembly fabrication shall be made with metal dies and jig fixtures to insure uniformity of size and shape.

3.19.2 Rivet connections. Rivet holes, in the sizes recommended by standard practice, shall be accurately punched or drilled, and shall have burrs removed. Rivets shall be driven with power tools. Rivet heads shall be full, neatly made, concentric with rivet holes, in full contact with the surface of the member. Excessive upsetting of rivets to fill holes will not be acceptable.

3.19.3 Bolted connections. Bolt holes shall be accurately formed. Washers or lockwashers or a combination of the two shall be provided on all bolts, studs and capscrews having straight threads. Self-locking nuts are acceptable in lieu of standard nuts and lockwashers. All fasteners shall be correctly tightened and shall have full thread engagement.

3.19.4 Welds. Welded joints shall be sound, smooth, free from pits, holes, fissures, rough projecting edges, and slag; and shall be thoroughly fused to the base metal.

3.19.5 Machine work. All parts shall be manufactured to gage through the use of correct jigs, fixtures or tape controlled machines, or any desired combination of these. Like parts shall be interchangeable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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shall be kept complete and available to the Government as specified in the contract or order.

4.1.2 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with the requirements of referenced specifications and standards. Any component or material which cannot be checked after final assembly shall be inspected in lots by sampling, for compliance with this specification and referenced specifications. Any deviation shall constitute a defect. The acceptable quality level (AQL) of each lot shall be 4.0 defects per 100 units. The inspection level shall be I of MIL-STD-105.

4.1.3 Disassembly inspection. Parts and components failing any test shall be disassembled in the presence of a Government representative. Each disassembled part shall be inspected in detail for compliance with this specification in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected.

4.2 Classification of inspection.

(a) First article inspection (see 4.3).

(b) Initial production inspection (see 4.4).

(c) Quality conformance inspection (see 4.5.1).

(d) Inspection comparison (see 4.5.2).
(e) Inspection of preparation for delivery (see 4.5.3).

4.3 First article inspection.

4.3.1 First article examination.

4.3.1.1 Pretest examination. The first article (a preproduction model) shall be examined for the defects marked "X" in column 1 of Table II. Presence of one or more defects shall be cause for rejection.

4.3.1.2 Post-test examination. Upon successful completion of all tests specified in 4.3.2, the preproduction model shall be examined for the defects marked "X" in column 2 of Table II. Presence of one or more defects shall be cause for rejection.

4.3.2 Preproduction tests. Upon successful completion of the examination specified in 4.3.1.1, the preproduction model shall be subjected to the tests marked "X" in column 1 of Table III. Failure of any test shall be cause for performing the inspection specified in 4.1.3 or shall be cause for rejection of the preproduction model for failure to meet the performance requirements of this specification.

4.4 Initial production inspection. An initial production truck will be selected at random by the Government from the first 10 trucks being produced from production tooling and will be examined as specified in column 3 of Table II and subjected to the tests marked "X" in column 2 of Table III, to determine conformance to the requirements of this specification. The inspection will be

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performed by the contractor unless otherwise specified (see 6.2). Acceptance of the initial production truck shall not exclude the remaining trucks from the quality conformance inspection and acceptance provisions specified in Section 4.

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TABLE II Examination Schedule

Pretest 1	Post test 2	In Production 3	Individual 4	Sample 5		Defects 6	Requirement Paragraph 7
X	-	-	X	-	101	Assembly not complete or not correct	Section 3
X	X	X	X	-	102	Components missing or not as specified	Section 3
X	-	X	X	-	103	Materials not as specified	3.2
X	-	X	X	-	104	Threads not in accordance with Handbook H28	3.2.3
X	-	X	X	-	105	Safety requirements not as specified-type E or type EE truck	3.3
X	-	X	X	-	106	Overload devices not provided with each circuit	3.5.1.3
X	-	X	-	X	107	Thermal relays for resistor housings not installed	3.5.1.4
X	-	X	X	-	108	Two (2) replaceable static discharge devices not supplied	3.5.1.5
X	-	X	X	-	109	Means of identification for conductors not provided	3.5.1.6
X	-	X	-	-	110	Vehicle not designed for 24 D.C. volt operation	3.5.2.1
X	-	X	-	X	111	Thermal relays for motors not installed	3.5.2.2
X	-	X	-	X	112	Contactors for speed controller not easily replaceable	3.5.2.4
X	-	X	-	X	113	Battery compartment or battery well dimensions not as specified	3.5.3.1
X	-	X	-	X	114	Battery connectors not as specified	3.5.3.2
X	-	X	-	X	115	Padlock for battery connector on EE trucks not provided	3.5.3.2
X	-	X	X	-	116	Ventilating holes for battery	

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						compartment or well not screened or louvered	3.5.3.3
X	-	X	-	X	117	Battery plug cannot be removed without unlocking cover	3.5.3.2
X	X	-	-	-	118	Drive assembly not capable of withstanding test torque without failure, fracture or deformation	3.5.4
X	X	X	X	-	119	Hydraulic system not as specified, does not function, or leaks	3.5.5
X	X	X	X	-	120	Contamination present in hydraulic reservoir return, and suction line	3.5.5
X	-	X	-	-	121	Piston rod not finished to 0.0005" chrome plate	3.5.5.1
X	X	X	-	X	122	Frame depth does not protect working parts	3.6.1
X	-	X	-	X	123	Fork dimensions and lift not as specified	3.6.2
X	-	X	-	X	124	Ball or tapered roller bearings not provided	3.6.4
X	X	X	X	-	125	Accessible grease fittings not provided for all rotating parts	3.6.5.1
X	-	X	X	-	126	Tires not as specified	3.6.5.2
X	-	X	-	X	127	Load wheels not made of conductive, nonsparking materials	3.6.7
X	X	X	X	X	128	Steering system not as specified, does not function, or leaks	3.6.9
X	-	X	-	-	129	Vehicle wheels do not provide for geometric steering	3.6.9
X	X	X	X	X	130	Brakes not as specified or do not function	3.6.10
X	-	X	-	X	131	Truck body does not completely cover drive unit assembly, steering mechanism and speed controller when these components are not housed in individual containers.	3.6.11
X	-	X	-	X	132	Easy access to all working parts is not provided	3.6.11
X	-	X	-	X	133	Operator's platform not as specified	3.6.11.1

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X	X	X	X	X	134	Controls and instrumentation not as specified or do not function correctly	3.7
X	-	X	-	X	135	All controls are not provided with heavy moisture and weather resistant markings	3.7
X	X	X	X	X	136	Directional speed controls and push button requirements do not function as specified	3.7.1
X	-	X	X	-	137	Lifting and lowering controls are not as specified	3.7.2
X	X	X	X	X	138	Instruments not located as specified or do not function correctly	3.7.3
X	X	X	X	-	139	Electromagnetic interference not as specified	3.9
X	X	X	X	X	140	Lubricants not as specified or not at correct level	3.10
X	X	X	X	X	141	Lubrication fittings not as specified, damaged, or subject to leakage	3.10.2
X	X	X	-	X	142	Lubrication fittings not accessible to a grease gun	3.10.2
X	-	X	-	X	143	Fungus and moisture resistance not as specified (is required)	3.11
X	-	X	X	-	144	Nameplate and registration data incorrect or not as specified	3.12, 3.13
X	-	X	X	X	145	Treatment and painting not as specified	3.14
X	-	X	X	-	146	Technical publications missing or not as specified	3.15
X	-	X	X	-	147	Repair parts and maintenance tools incorrect or missing	3.16
X	X	X	X	X	148	Workmanship not as specified	3.19
X	X	X	X	X	149	Incorrect fit at joints, seams, or edges	3.19
X	X	X	X	X	150	Welding not as specified	3.19.4

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TABLE III Test Schedule

Production Model	Production Initial	Individual	Sample	Test	Test Paragraph	Requirement Paragraph	Test Method Number
1	2	3	4	5	6	7	8
X	X	X	X	Maximum load test	4.6.1	3.8.1	
X	-	-	X	Shock load	4.6.2	3.8.2	
X	X	-	X	Turning radius	4.6.3	3.8.4	
X	-	-	X	Pallet entry and withdrawal	4.6.4	3.8.3	
X	X	-	X	Speed -	4.6.5	3.8.5	
X	X	-	-	Slope ascension	4.6.6	3.8.6	
X	X	X	X	Lifting & lowering speed	4.6.7	3.8.8	
X	X	-	-	Stability	3.6.8	3.8.12	
X	X	X	X	Leakage	4.6.9	3.8.10	
X	X	X	X	Brakes	4.6.10	3.8.11	
X	X	-	-	Endurance test	4.6.11	3.8.9	
X	X	-	-	Operator's platform test	4.6.12	3.6.11.1	
X	X	-	X	Drive unit mounting	4.6.13	3.6.8	
X	-	-	-	Operating temperature test	4.6.14	3.5.1.4	2
X	-	-	-	Electromagnetic interference	4.6.15	3.9	
X	-	-	-	Power consumption	4.6.16	3.4.2	4
X	-	-	-	Maintainability	4.6.17	3.4	3
X	X	-	-	Drive unit	4.6.18	3.5.4	1

4.4.1 Inspection failure. Failure of the initial production truck to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production truck and shall be cause for refusal by the Government to continue acceptance of

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production trucks until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on truck previously accepted and produced under the contract. Any deficiencies found as a result of the initial production testing will be considered prima facia evidence that all trucks accepted prior to the completion of initial production tests are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Individual examination. After successful completion of all tests specified in 4.5.2 each truck shall be examined for the defects marked "X" in column 4 of Table II. Presence of one or more defects shall be cause for rejection.

4.5.2 Individual tests. Each truck shall be subjected to the tests marked "X" in column 3 of Table III. Failure of any tests shall be cause for performing the inspection specified in 4.1.3.

4.5.3 Sampling.

4.5.3.1 Unit of product. A unit of product is defined as a fully serviced truck.

4.5.3.2 Lot. A lot shall consist of not more than 50 units of product. Lots shall be formulated from trucks which have successfully completed the individual examination and individual tests as specified in 4.5.1 and 4.5.2 respectively.

4.5.3.3 Sample tests. One sample shall be selected from each lot and shall be subjected to the tests marked "X" in column 4 of Table III. Any sample failing any test shall be cause for rejection of the lot.

4.5.3.4 Sample examination. After successful completion of the tests as specified in 4.5.3.3 each sample shall be examined for the defects marked "X" in column 5 of Table II. Presence of one or more defects shall be cause for rejection of the lot.

4.5.4 Inspection comparison. The Government may select trucks at any time during the contract production period and subject these trucks to the examination specified in columns 1 and 4 of Table II and to the tests specified in column 1 of Table III to reveal any manufacturing deficiencies. The inspection will be performed by the Government at a site selected by the Government. Trucks will be selected at random from the previously selected preproduction models, the initial production model, and those trucks tested in 4.5.3. Rehabilitation shall be at Government expense.

4.5.4.1 Inspection failure. Failure of any truck when examined and tested as specified in 4.6 may be considered cause for refusal by the Government to continue acceptance of trucks until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection.

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4.5.5 Examination of preparation of delivery. An examination will be made to determine that requirements for preservation, packaging, packing and marking are in compliance with MIL-STD-162.

4.6 Test procedures.

4.6.1 Maximum load. The truck under test shall be placed on a smooth, dry, level concrete floor with forks elevated to maximum lift height. A uniformly distributed load of 8,000 pounds (3630 kg) for size I, 12,000 pounds (5444 kg) for size II, and 20,000 pounds (9070 kg) for size III, shall be placed on the test unit. The truck shall remain under test load for a period of 1 hour, during which time the drift shall not exceed 1/4 inch (6 mm) . After the 1 hour period has elapsed, the load shall be removed and the truck inspected. Any evidence of permanent deformation, skewing, or failure shall be sufficient evidence of failure to comply with the specification.

4.6.2 Shock load. The pressure relief valve shall be made inoperative for this test. A uniformly distributed load of 6,000 pounds (2722 kg) for size I, 9,000 pounds (3633 kg) for size II, and 15,000 pounds (6800 kg) for size III, shall be placed on the test unit. The truck under test shall be driven off a 1-inch (25 mm) high platform 5 times at maximum speed. Joints shall then be tightened to compensate for settling. The truck under test with forks at maximum height shall be placed on a surface plate. The following heights shall be measured and recorded and the points of measurement marked on the truck: (a) front end of frame, (b) front pivot point of raising and lowering mechanism, (c) top of forks at heel on both sides, (d) rear pivot point of raising and lowering mechanism on both sides. The previous test shall be repeated by driving the loaded truck off a 1-inch high platform 10 times. The truck shall be replaced on surface plate. Heights to the referenced points previously described herein shall be measured and recorded. A difference in height between the first and second measurements at any reference point exceeding 1/8 inch (3 mm) or any evidence of permanent deformation, skewing, or failure shall constitute non-compliance and shall be cause for rejection. Wheel bearings shall be inspected to assure that they are free from distortion or binding.

4.6.3 Turning radius. The truck with rated load equally distributed on the forks shall be placed on a smooth, dry, level concrete surface. The truck shall be operated to travel in a circular path to determine compliance with the requirements of 3.8.4. Any evidence of non-compliance shall be cause for rejection of the truck.

4.6.4 Pallet entry and withdrawal. An empty double faced standard military pallet shall be placed with its front edge on a reference line, on a smooth dry concrete floor. The truck under test shall be fully engaged into the pallet. This distance from the reference line to some reference point on the pallet edge shall be measured. This test shall be repeated 10 times. The pallet edge shall be placed on a reference line, and the forks shall be fully engaged into the pallet; forks shall be raised to maximum lift height and then lowered complete. The truck shall be withdrawn from the pallet. The distance from the reference line to the same reference point on the pallet edge shall be measured. The test shall be repeated 10 times. If average distance through which the pallet moves, either in a forward or backward direction exceeds 6 inches (152 mm), as specified in 3.8.3, it shall be sufficient evidence of failure to comply with this specification.

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4.6.5 Speed. Truck shall be accelerated, carrying rated load, through a 50 foot (15 m) approach, timed through a 88 foot (27 m) test length, and then brought to a stop. (This 88 foot (27 m) length is used to simplify calculations 88 feet (27 m) per minute is equal to 1 mile per hour) (1.6 kilometer per hour). Three runs shall be made in both the forward and reverse direction in high speed. The data is recorded and the average speed determined. Evidence of the truck's failure to comply with the requirements of 3.8.5 shall be sufficient cause for rejection of the truck under test.

4.6.6 Slope ascension. The unit under test carrying rated load shall be accelerated along a horizontal surface and onto a 15 percent ramp and brought to a dead stop (see 3.8.6). The traction motor shall then be energized and the truck shall accelerate the ramp. Failure to meet this requirement shall be cause for rejection of the truck under test.

4.6.7 Lifting and lowering speed. Rated load shall be placed on the forks with truck on a level surface. The hoist motor shall be energized and the load lifted to full lift height. The time in seconds shall be taken by a timing device from the instant forks begin to rise to the end of full lift height. This test shall be repeated at least 5 times and all data tabulated and averaged. The time required to raise rated load to maximum lift height shall not exceed the limits specified in 3.8.8. The lower control shall be actuated and load stopped at random heights by deactivation of the lowering control. Nonconformance with requirement of 3.8.8 shall be cause for rejection.

4.6.8 Stability test: The truck shall be subjected to the following stability test. The truck shall be required to lift rated load to maximum elevated height and propelled in a clockwise direction through 360 degrees after which the brakes shall be applied. The load shall then be lowered, raised again and the truck propelled 360 degrees in a counterclockwise direction and brought to a complete stop. This test shall be accomplished 3 times in each direction after which 3 cycles in each direction shall be completed without the load. Failure to meet this requirement shall be cause for rejection of the truck under test.

4.6.9 Leakage. After completion of the above tests, the hydraulic system shall be inspected. The observation of any oil or grease in the vicinity of the reservoir or the hydraulic lines shall be evidence of failure to comply with this specification.

4.6.10 Brakes. Place rated load on forks and bind to forks. Raise forks to maximum lift height and accelerate truck in a forward direction at maximum speed. The truck shall be stopped quickly and smoothly without grabbing. Measure distance required to stop truck after the brakes are applied. Evidence of failure to stop truck within 5 feet will be sufficient cause for rejection of the truck under test.

4.6.11 Endurance test. When a preproduction model is required, the truck shall be subjected to 500 cycles of the endurance test. One cycle of the endurance test shall be as outlined herein. The truck shall be required to lift rated load to maximum elevated height and propelled in a clockwise direction through 360 degrees after which the brakes shall be applied. The load shall then be lowered, raised again and the truck propelled 360 degrees in a counterclockwise direction and brought to a complete stop. A five minute rest period may be allowed between each 25 cycles completed. Next 25 cycles are to be run without

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load. At the end of 500 cycles, the rated load shall be lifted to maximum lift height and held for a period of 5 minutes. Drift of the load shall not exceed one-half inch per minute. Failure to meet the requirements of the test shall be cause for rejection of the truck under test.

4.6.12 Operator's platform test. The truck or the platform shall be tested to determine compliance with 3.6.11.1. Evidence of failure shall be cause for rejection of the truck.

4.6.13 Drive unit mounting. When two drive units are used, mounting shall have sufficient articulation for one wheel to negotiate a 1-inch obstacle block while the other wheel remains in contact with the floor. Failure to meet the requirements of 3.6.8 shall be cause for rejection of the truck under test.

4.6.14 Operating temperature test. The type EE truck shall be operated at low speed over a level surfaced test course in a facility where the ambient temperature shall be between 95 degrees F (35 degrees C) and 100 degrees F (38 degrees C). The test course and method of negotiating the course shall be in accordance with the operating temperature test of Test Method No. 2. The truck shall be provided with sufficient heat sensing devices located at critical heat areas on the outside surfaces of the drive motor and resistor housing, and a temperature recording instrument to continuously record these temperatures during the test run. This test shall be run with continuous operation of truck until temperatures being recorded have stabilized. Temperature shall be considered as being stabilized when the temperature being recorded does not increase (remains constant) over a minimum period of one hour during continuous operation of the truck. Failure to meet requirements of paragraph 3.5.1.4 shall be cause for rejection of the truck.

4.6.15 Electromagnetic interference. Test to determine compliance with 3.9 shall be performed in accordance with the requirements of the applicable specification. Failure to meet requirements will be cause for rejection.

4.6.16 Power Consumption. Test the truck in accordance with Test Method Number 4. Failure to conform with requirements of the test method shall constitute failure of this test.

4.6.17 Maintainability. Test the truck in accordance with Test Method Number 3. Failure to conform with requirements of the test method shall constitute failure of the test.

4.6.18 Drive unit. Test the drive unit in accordance with Test Method Number 1. Failure to conform with requirements of the test method shall constitute failure of this test.

5. PREPARATION FOR DELIVERY.

5.1 Preservation, packaging and packing. Preservation, packaging, and packing shall be level A, B, or C as specified in accordance with MIL-STD-162 (see 6.2).

5.2 Marking. In addition to any special marking required (see 6.2) shipment shall be marked in accordance with MIL-STD-129.

6. NOTES

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6.1 Intended use. The electric-powered, low-lift, stand-up rider type pallet truck with forks is intended for moving general supplies or cargo. It is also used in handling propellants, ammunition and other ordnance applications in and around warehouses, loading platforms, ordnance depots, ammunition depots, docks and on-board ships within the military supply and weapons support system. Trucks are intended for operation over paved or other hard surfaces for short distances. Also type EE trucks are for use where continuous operation is required at slow speeds such as inching with minimum temperature rise of electrical components.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Type and size required (see 1.2)
(c) When preproduction model is required (see 3.1.2)
(d) When power consumption test is required (see 3.4.1)
(e) When batteries are other than Government furnished (see 3.5.3.1)
(f) When battery compartment or well is required (see 3.5.3.3)
(g) When drive assembly test is required (see 3.5.4)
(h) Length of forks required (see 3.6.2)
(i) When pallet positioner is required (see 3.6.3)
(j) When fire extinguisher is required (see 3.7.5)
(k) When radio interference suppression, other than that specified, is required (see 3.9)
(l) Types of military lubricants required (see 3.10.1)
(m) When fungus and moisture resistance is required (see 3.11)
(n) When registration data is other than required (see 3.13)
(o) When additional distribution of technical manuals is required (see 3.15)
(p) Repair parts and maintenance tools required (see 3.16)
(q) Level of preservation, packaging and packing required (see 5.1)
(r) Special marking required in addition to MIL-STD-129 (see 5.2)
(s) When initial production inspection shall be conducted by other than the contractor (see 4.4)

APPENDIX

FOR

TRUCK, PALLET, POWERED, ELECTRIC,

LOW-LIFT, STAND-UP RIDER TYPE

7. SCOPE

The test methods contained within this appendix are to determine whether trucks procured under this specification conform to the requirements set forth.

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TEST METHOD NO. 1

DRIVE TRAIN STATIC TORQUE TEST

1. Test Course:

Shop area.

2. Test Apparatus:

(a) Torque arm.
(b) Tension dynamometer.

3. Test Procedure:

(a) Transmission shall be placed in lowest gear range.

(b) Lock both drive wheels to prevent turning.

(c) If transmission is the hydraulic power operated type, increase hydraulic pressure or lock clutch by external means to prevent slippage.

(d) Clamp a torque arm of (L) feet to the input shaft of the transmission or in case of power operated to the output shaft of torque converter.

(e) Obtain from manufacturer's published data, the bare engine torque (T) foot pounds of the engine. If drive includes hydraulic torque converter obtain also the stalled torque converter ratio (R).

(f) Apply to the torque arm, a force equal to $\frac{1.5T}{L}$ for mechanical transmission; apply $\frac{1.5TR}{L}$ for trucks equipped with hydraulic converter.

(g) Disassemble transmission and examine for failures, fractures, or permanent distortion and deformation.

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TEST METHOD NO. 2

TEMPERATURE STABILIZATION

1. Test Course:

   (a) Shall be in accordance with Figure 1.

2. Test Apparatus:

   (a) Thermocouple(s).
   (b) Pyrometer.
   (c) Oven.
   (d) Thermometer

3. Test Procedure:

   (a) Remove thermal switches and place in oven. Slowly INCREASE temperature until switches open. Read and record temperatures at which thermal switches open.

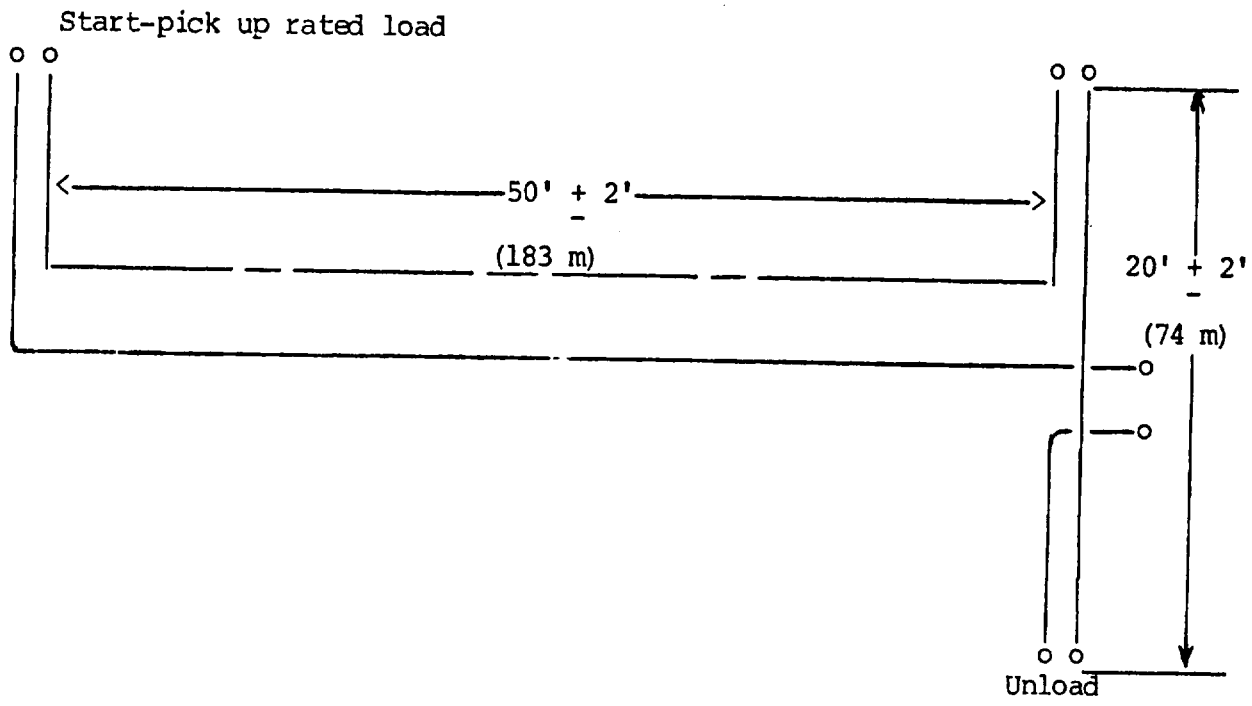
   (b) Reinstall thermal switches on truck and install thermocouple(s) adjacent to thermal switches. Additional thermocouple shall be submerged in the hydraulic reservoir.

   (c) When required, place truck in facility where ambient temperature can be maintained.

   (d) Operate truck continuously at low speed over the test course. Read and record temperatures including ambient at intervals not more than 15 minutes. Continue operation on test course until all critical temperatures stabilize for two consecutive readings.

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FIGURE 1 (TEST METHOD NO. 2)



Reverse, Low Speed, W/Rated Load
Forward, Low Speed, W/Rated Load
Reverse, Low Speed, W/O Rated Load
Forward, Low Speed, W/O Rated Load

Note: On alternate laps, the load should be picked up where it was previously deposited and redeposited where it was previously picked up. The load shall be raised to full lift height before it is deposited.

TEST COURSE FOR ELECTRIC (SPARK ENCLOSED) FORK LIFT TRUCK

(MOTOR AND RESISTOR HOUSING HEAT STABILIZATION)

FIGURE 1 (TEST METHOD NO. 2)

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TEST METHOD NO. 3

MAINTAINABILITY

1. Test Course:

Shop area of sufficient size to permit unobstructed surveillance of maintenance.

2. Test Apparatus:

(a) Common handtools.
(b) Special tools (if furnished with truck).

3. Test Procedure:

(a) Examine truck for accessibility of major assemblies.

(b) Examine position of drains with respect to accessibility.

(c) Using the tools, demonstrate that major assemblies are accessible for repair and maintenance.

(d) Demonstrate both the accessibility of drains and the path of discharge of lubricants by activating drainage controls.

(e) Perform and time each operation on the maintenance operations list (see 3.4.1).

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TEST METHOD NO. 4

POWER CONSUMPTION TEST

1. Test Course:

Dry, level area of sufficient size to perform test described below.

2. Test Apparatus:

(a) Stop watch.

3. Test Procedure:

(a) The truck shall be cycled continuously for 7-1/2 hours with one battery only (battery shall not be charged during the test).

(b) Each cycle detailed below shall take 2-3/4 +/- 1/4 minutes.

(c) Cycles shall be timed and recorded.

(d) One cycle consists of the following operations on the test course.

(1) Pick up capacity load "A" at area "X" and place it at area "Z".

(2) Back up to area "Y".

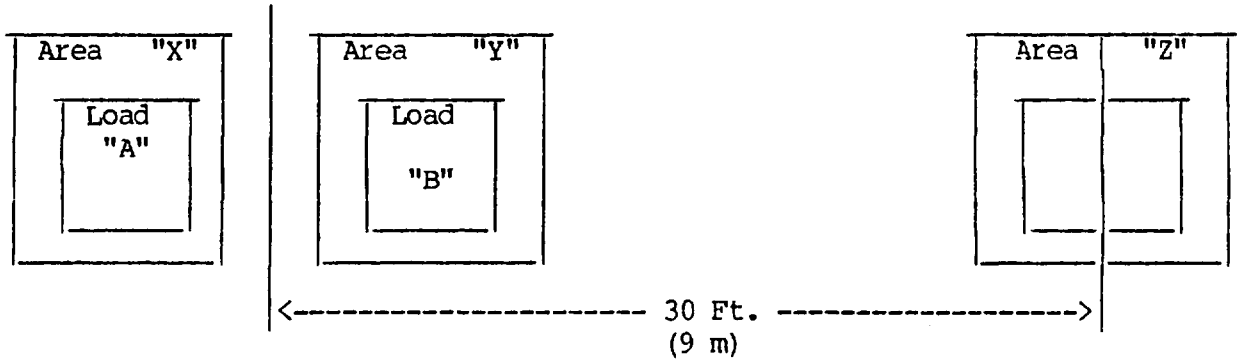
(3) Pick up capacity load "B" at area "Y" and place it next to load "A" at area "Z".

(4) Back out and drop forks.

(5) Raise forks, pick up load "B" and place it at area "Y".

(6) Back out to area "Z".

(7) Pick up load and place it at area "X".



(e) Record number of cycles per hour.
(f) Observe that truck negotiated test course for 7-1/2 hours.

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Custodians:

Army	- ME
Navy	- SA
Air Force	- None

Review Activities:

Army	- ME, MU
Navy	- SA, SH

User Activities:

Army	- SM
Navy	- OS

Preparing Activity:

Navy - SA

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